PACLAND · Drug discovery

The verification discipline applied to computational drug discovery — a plain-English brief

1 · What we are measuring

Drug discovery is a search over chemical space (~10^{60} molecules) with scarce ground truth: an assay exists only for the molecules you have already tested. The discipline asks how much of a screening decision is verified, versus predicted.

2 · The verification deficit

- 4 of 10 benchmark endpoints are measured + computable + validated — a certificate radius of 68%.
- 0 of 5 endpoints for a novel molecule are validated — the hit you would synthesize is extrapolation, not verification.

3 · The constraint guard-suite

- 4 guards, each mutation-tested: net charge, valence bounds, Lipinski (rule of five), synthesizability.

The field has run these 'forbidden-state' filters for 25 years — PAINS, REOS, Lipinski, RDKit sanitization — as boolean heuristics. The guard-suite wraps them as a pass/fail test a model must pass, with a seeded fault proving each guard fires.

4 · Why it matters

The field's own 2013 census (Zhu et al.) found 52% of reported virtual-screening hits fail REOS and 18% are PAINS artifacts, and AlphaFold's confidence score (pLDDT) is not calibrated. 'Verified' hits are, in large part, artifacts — and the discipline makes that a machine-checked verdict.

5 · Reproduce

pacland drugs · docs/drugs-verification-deficit.md